

# Applied Optimization

## Exercise 7 - L-BFGS

November 19, 2025

### Secant Equation (0.5 pt)

The secant equation is:

$$B_{k+1}s_k = y_k,$$

where  $s_k = x_{k+1} - x_k$ ,  $y_k = \nabla f_{k+1} - \nabla f_k$ , and  $B_{k+1}$  is an approximation of the Hessian  $\nabla^2 f_{k+1}$ .

Prove that the secant equation is valid to find the next hessian approximation  $B_{k+1}$

### Solution:

Start by assuming a quadratic model

$$m_k(v) = f_k + \nabla f_k^T v + \frac{1}{2} v^T B_k v,$$

from which the optimal search direction  $v_k$  can be computed by computing  $\nabla m_k(v) = \nabla f_k + B_k v$ , setting it to zero and solving for  $v$  to obtain  $v_k = -B_k^{-1} \nabla f_k$ . This would give, for a given step  $t_k$ , the next iterate  $x_{k+1} = x_k + t_k v_k$ .

For our hessian approximation for the next iterate  $B_{k+1}$ , we require the gradients of the function and model to coincide in the following way:

$$\nabla m_{k+1}(0) = \nabla f_{k+1} \quad \nabla m_{k+1}(-t_k v_k) = \nabla f_k,$$

which means, the model gradient evaluated at  $v = 0$  corresponds to the function gradient at  $x_{k+1}$  and the model gradient at  $v = -t_k v_k$  to the function gradient at the previous iterate  $x_k$ .

$$\nabla m_{k+1}(0) = \nabla f_{k+1} + B_{k+1}(0) = \nabla f_{k+1}$$

and

$$\nabla m_{k+1}(-t_k v_k) = \nabla f_{k+1} + B_{k+1}(-t_k v_k) = \nabla f_k$$

from the expression for the next iterate we know that  $x_{k+1} - x_k = t_k v_k$ , thus

$$\nabla f_{k+1} - \nabla f_k = B_{k+1}(x_{k+1} - x_k),$$

from which follows the secant equation substituting  $y_k$  and  $s_k$ .

## Curvature Condition (0.5 pt)

Show that the curvature condition  $s_k^T y_k > 0$  is required for the Quasi-Newton method. Prove that the curvature condition is satisfied when the line search algorithm for the Wolfe conditions is used.

### Solution:

The Hessian approximation needs to be symmetric and positive semidefinite because that's what we need for Newton's method. We can start from the secant equation,

$$B_{k+1} s_k = y_k$$

and multiply both sides by  $s_k^T$  to get

$$s_k^T B_{k+1} s_k = s_k^T y_k > 0$$

This gives us the curvature condition since we require  $B_{k+1}$  to be p.d. In summary, the curvature condition is necessary for the Quasi-Newton method.

To show that  $s_k^T y_k > 0$  follows from the line search that satisfies the Wolfe conditions, substitute  $s_k$  and  $y_k$  and use  $\nabla f_{k+1}^T p_k \geq \beta \nabla f_k^T p_k$  with  $\beta \in [0, 1[$ . Take for convenience  $s^T y = y^T s$ .

$$\begin{aligned} y_k^T s_k &= (\nabla f_{k+1} - \nabla f_k)^T (\alpha_k p_k) \\ &= \alpha_k (\nabla f_{k+1}^T p_k - \nabla f_k^T p_k) \\ &\geq \alpha_k (\beta \nabla f_k^T p_k - \nabla f_k^T p_k) \\ &= \underbrace{\alpha_k}_{>0} \underbrace{(\beta - 1)}_{<0} \underbrace{\nabla f_k^T p_k}_{<0} > 0 \end{aligned}$$

$\alpha_k > 0$  follows from the step length constraint for line search, the slope  $\nabla f_k^T p_k$  is negative if  $p_k$  is a descent direction, and  $\beta - 1 < 0$  since  $\beta < 1$ .